

**8051 MICROCONTROLLER BASED PROJECT REPORT ON
Electronic Voting Machine using Arduino UNO**

This project report is submitted to

**Yeshwantrao Chavan College of Engineering
(An Autonomous Institution Affiliated to Rashtrasant Tukdoji
Maharaj Nagpur University)**

of
Bachelor of Engineering in Electronics Engineering

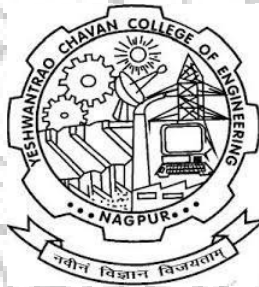
By

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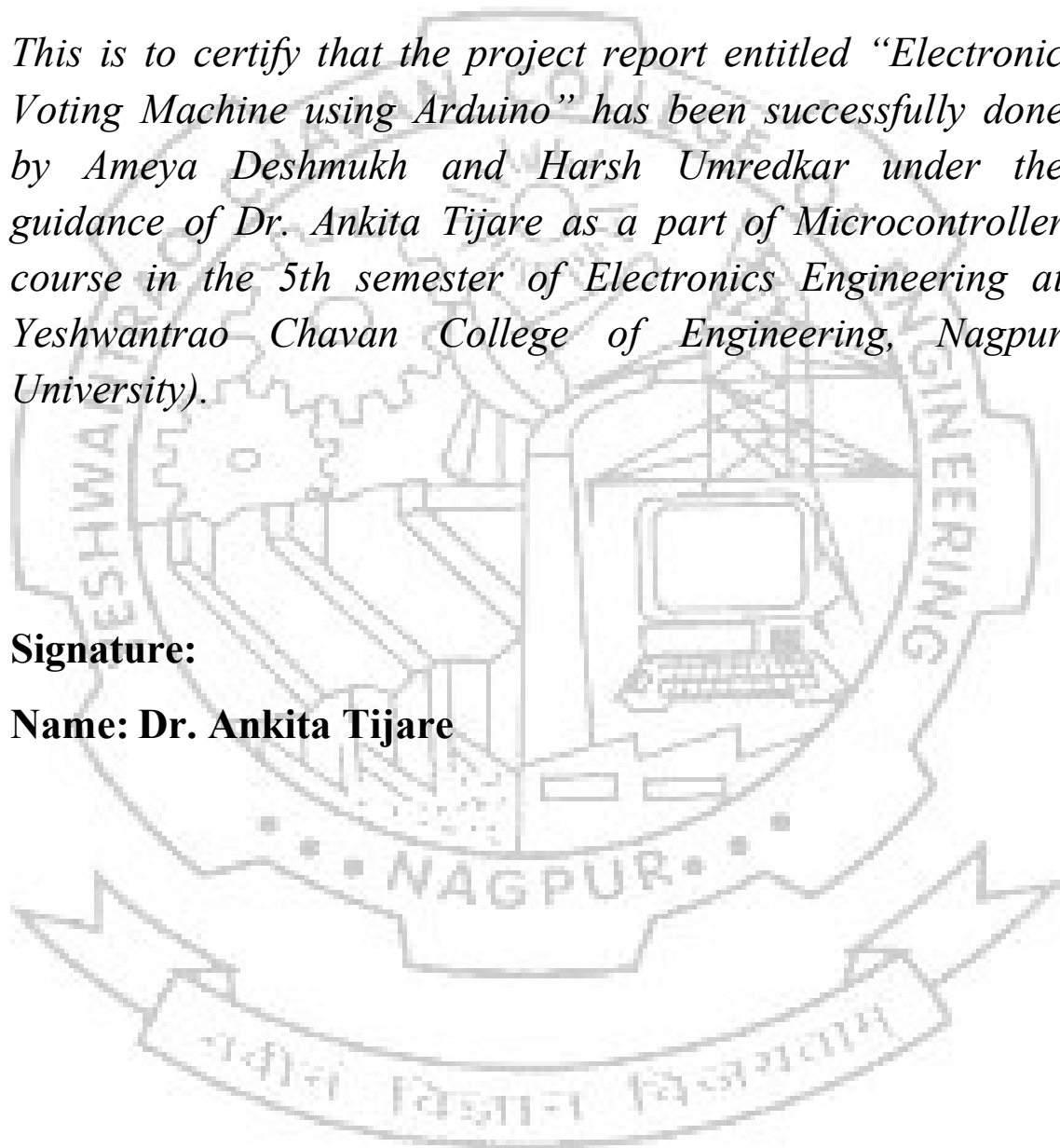
May 2025

CERTIFICATE

This is to certify that the project report entitled “Electronic Voting Machine using Arduino” has been successfully done by Ameya Deshmukh and Harsh Umredkar under the guidance of Dr. Ankita Tijare as a part of Microcontroller course in the 5th semester of Electronics Engineering at Yeshwantrao Chavan College of Engineering, Nagpur University).

Signature:

Name: Dr. Ankita Tijare

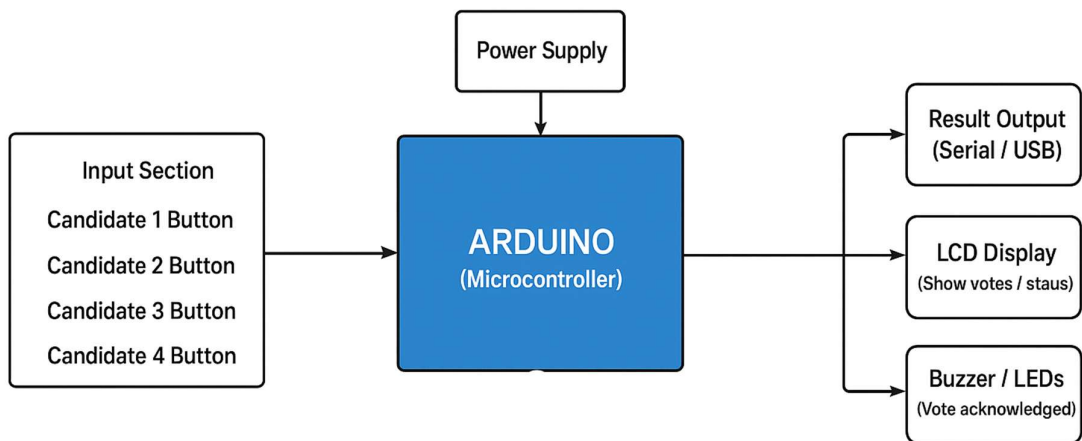


Problem Statement

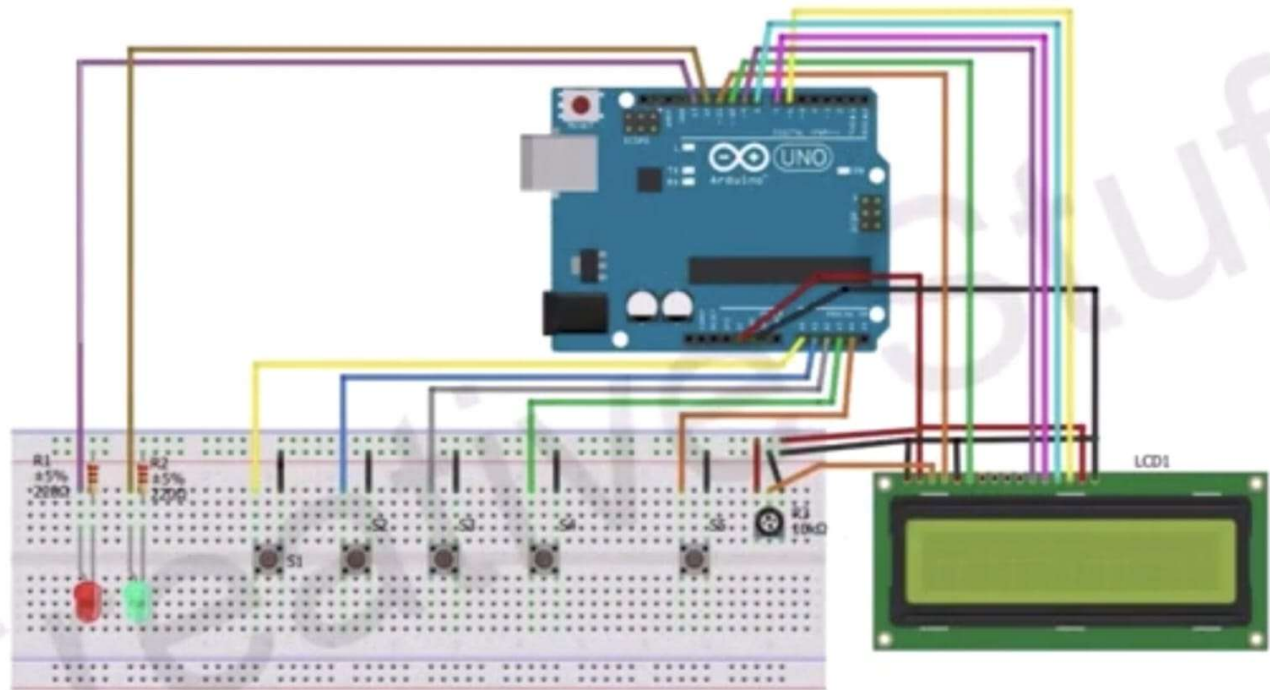
Design and develop an **Electronic Voting Machine (EVM)** using **Arduino** that enables secure, efficient, and user-friendly voting for multiple candidates. The system should allow each voter to cast a single vote through push buttons assigned to different candidates and automatically display the final results after the voting session ends. This project aims to minimize human error, ensure vote accuracy, and provide a low-cost, reliable solution suitable for small-scale elections such as schools, colleges, and community organizations.

Block Diagram of the System

Block Diagram — Electronic Voting Machine (Arduino)

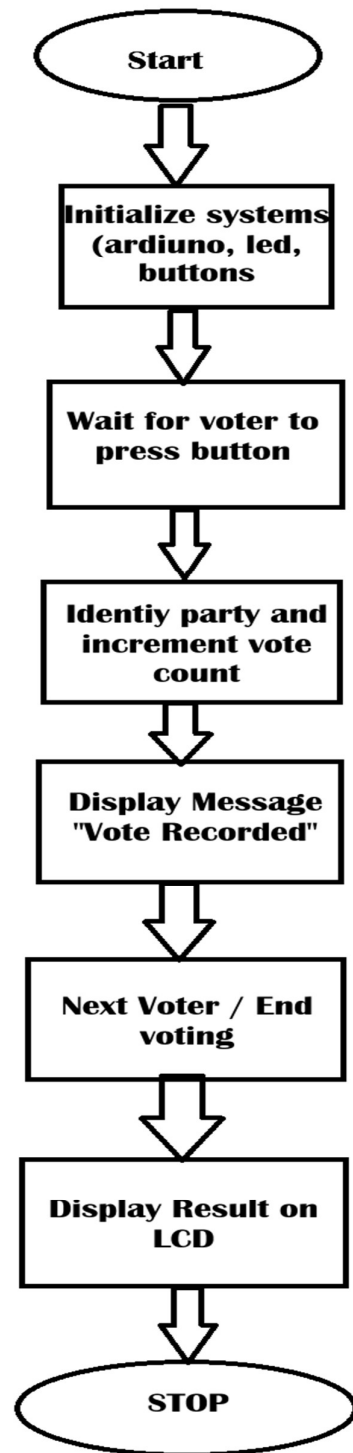


Circuit Diagram/Pin Connection



Electronics Voting Machine

Flowchart / Algorithm



C language Code

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(11,10,9,8,7,6);

#define sw1 A0
#define sw2 A1
#define sw3 A2
#define sw4 17
#define sw5 18
#define buzzer 2

int vote1=0;
int vote2=0;
int vote3=0;
int vote4=0;

void setup() {
  pinMode(sw1, INPUT);
  pinMode(sw2, INPUT);
  pinMode(sw3, INPUT);
  pinMode(sw4, INPUT);
  pinMode(sw5, INPUT);
  pinMode(13, OUTPUT);
  pinMode(12, OUTPUT);
  pinMode(buzzer, OUTPUT);

  lcd.begin(16, 2);
  lcd.setCursor(0,0);
  lcd.print(" MCI PROJECT");
  lcd.setCursor(0,1);
  lcd.print(" ROLL NO 32 & 49");
  delay(3000);
  lcd.clear();
  lcd.print(" VOTING MACHINE ");
  lcd.setCursor(0,1);
  lcd.print(" LCD DISPLAY");
  delay(3000);

  digitalWrite(sw1, HIGH);
  digitalWrite(sw2, HIGH);
  digitalWrite(sw3, HIGH);
  digitalWrite(sw4, HIGH);
  digitalWrite(sw5, HIGH);

  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("BJP");
  lcd.setCursor(4,0);
  lcd.print("INC");
```

```

    lcd.setCursor(8,0);
    lcd.print("AAP");
    lcd.setCursor(12,0);
    lcd.print("OTH");
}

void loop() {
    lcd.setCursor(0,0); lcd.print("BJP");
    lcd.setCursor(1,1); lcd.print(vote1);
    lcd.setCursor(4,0); lcd.print("INC");
    lcd.setCursor(5,1); lcd.print(vote2);
    lcd.setCursor(8,0); lcd.print("AAP");
    lcd.setCursor(9,1); lcd.print(vote3);
    lcd.setCursor(12,0); lcd.print("OTH");
    lcd.setCursor(13,1); lcd.print(vote4);

    if(digitalRead(sw1)==0) {
        vote1++;
        buzz();
        digitalWrite(12,HIGH);
        while(digitalRead(sw1)==0);
        digitalWrite(12,LOW);
        delay(1000);
    }

    if(digitalRead(sw2)==0) {
        vote2++;
        buzz();
        digitalWrite(12,HIGH);
        while(digitalRead(sw2)==0);
        digitalWrite(12,LOW);
        delay(1000);
    }

    if(digitalRead(sw3)==0) {
        vote3++;
        buzz();
        digitalWrite(12,HIGH);
        while(digitalRead(sw3)==0);
        digitalWrite(12,LOW);
        delay(1000);
    }

    if(digitalRead(sw4)==0) {
        vote4++;
        buzz();
        digitalWrite(12,HIGH);
        while(digitalRead(sw4)==0);
        digitalWrite(12,LOW);
        delay(1000);
    }

    if(digitalRead(sw5)==0) {
        digitalWrite(13,HIGH);
        int vote = vote1 + vote2 + vote3 + vote4;
    }
}

```



```

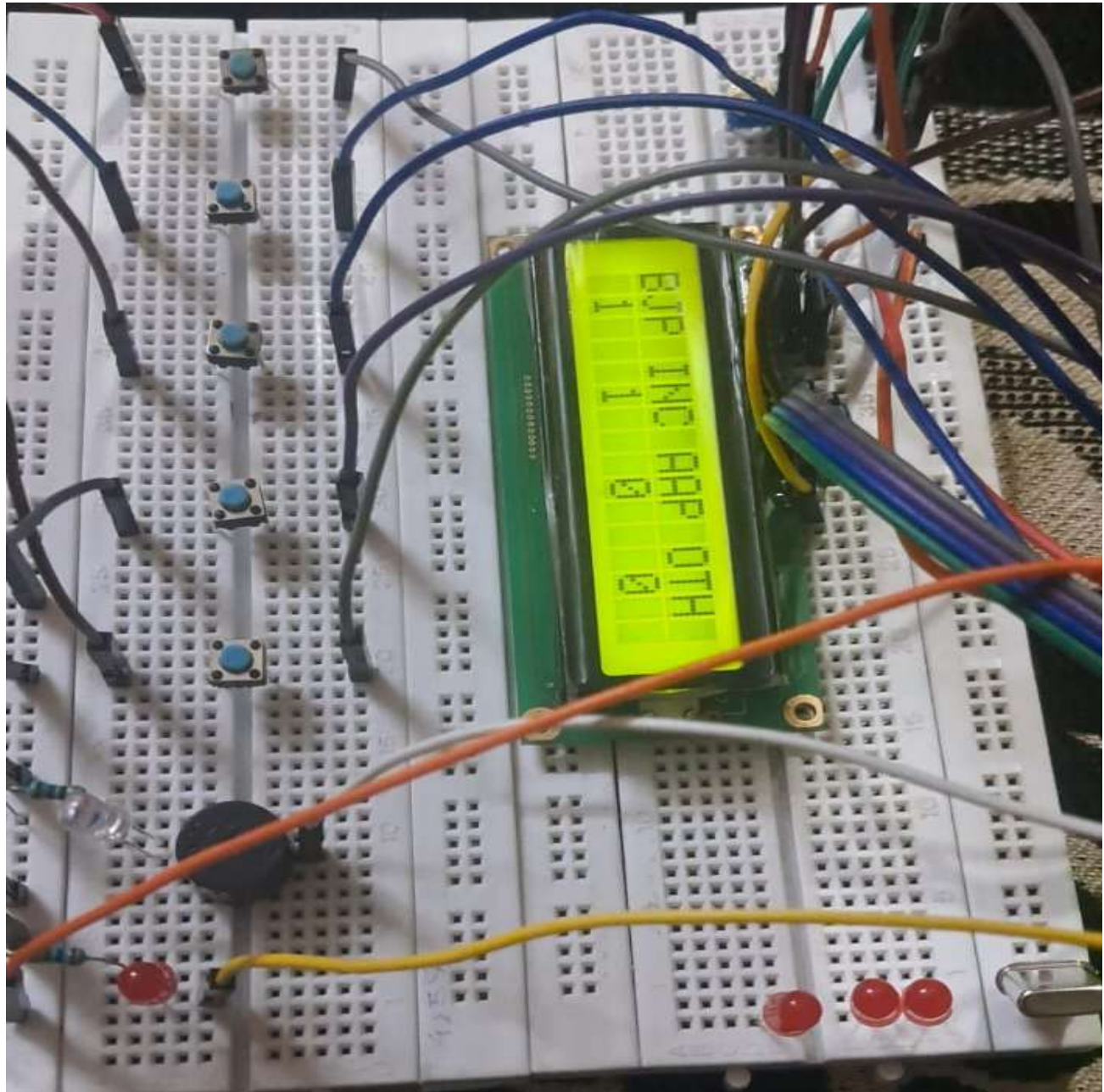
if(vote) {
    lcd.clear();
    if(vote1 > vote2 && vote1 > vote3 && vote1 > vote4) lcd.print("BJP Wins");
    else if(vote2 > vote1 && vote2 > vote3 && vote2 > vote4) lcd.print("INC Wins");
    else if(vote3 > vote1 && vote3 > vote2 && vote3 > vote4) lcd.print("AAP Wins");
    else if(vote4 > vote1 && vote4 > vote2 && vote4 > vote3) lcd.print("OTH Wins");
    else {
        lcd.print(" Tie Up Or ");
        lcd.setCursor(0,1);
        lcd.print(" No Result ");
    }
    delay(5000);
    lcd.clear();
} else {
    lcd.clear();
    lcd.print("No Voting....");
    delay(5000);
    lcd.clear();
}

vote1 = vote2 = vote3 = vote4 = 0;
digitalWrite(12, HIGH);
digitalWrite(13, LOW);
}
}

void buzz() {
    digitalWrite(buzzer, HIGH);
    delay(200);
    digitalWrite(buzzer, LOW);
}

```

Expected Output/Results



Applications and Limitations

Applications

1. **Educational Institutions:**
Used for student council or class representative elections in schools and colleges.
2. **Clubs and Organizations:**
Suitable for conducting internal polls or elections in clubs, societies, or small communities.
3. **Local or Small-Scale Elections:**
Can be used in housing societies, offices, or NGOs for quick and fair voting.
4. **Prototyping and Research:**
Helps students and researchers understand digital voting systems and microcontroller-based automation.
5. **Exhibitions and Demonstrations:**
Useful as a demonstration model to showcase the concept of electronic voting at science fairs or tech events.

Limitations

1. **Limited Security:**
Basic Arduino setup lacks encryption or authentication, making it less secure for official elections.
 2. **Limited Voter Capacity:**
Designed for small-scale use — cannot handle large numbers of voters or candidates.
 3. **No Biometric Verification:**
Cannot verify voter identity, so the same person could vote multiple times unless externally monitored.
 4. **No Data Backup:**
Votes are stored temporarily in Arduino's memory and can be lost if power fails.
 5. **Manual Result Reset:**
System needs manual resetting before a new election can start; not fully automated.
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